

VINOD SURESH

AI/ML Engineer | Robot Learning Research | ML Systems

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PROFESSIONAL SUMMARY

AI/ML engineer and Technical Lead with 9+ years of software engineering and production machine-learning experience, including Senior Machine Learning Engineer work at Microsoft. Doctoral research focuses on robot manipulation, VQ-VAE action tokenization, and cross-embodiment skill transfer. Builds simulation-based evaluation and runtime reliability tools alongside Python/PyTorch models, distributed training pipelines, and cloud ML systems. Combines hands-on engineering, structured experimentation, model observability, and technical leadership.

CORE SKILLS

Robot learning and simulation Robotic manipulation, reinforcement learning, trajectory modeling, cross-embodiment transfer, RLBench, MuJoCo, policy evaluation, runtime supervision, failure analysis and recovery.

Models and perception Python, PyTorch, TensorFlow, VQ-VAE, Transformers, graph neural networks, representation learning, optimal transport, contrastive learning, OpenCV, YOLOv8 and multimodal AI.

ML systems and evaluation Distributed training, Kubernetes, Kubeflow, MLflow, Docker, Azure, AWS, Spark, SQL, FastAPI, Neo4j, experiment tracking, ablations, statistical testing, drift detection and CI/CD.

SELECTED ROBOTICS AND AI RESEARCH

POLARIS | Doctoral Research at Walsh College

- Developing transferable policy representations for manipulation across Franka Panda, UR5, and custom 4-DOF embodiments, addressing morphology mismatch and heterogeneous action spaces.
- Architecture combines VQ-VAE action tokenization, morphology encoding, optimal-transport and contrastive/cycle-consistency alignment, and Transformer-based morphology-conditioned policies.
- Built RLBench-based training and evaluation workflows; studies zero-shot transfer, reconstruction quality, trajectory similarity, and cross-robot generalization.

Entropy-Adaptive Tokenization | Robot Trajectory Representations

- Built a manipulation-data pipeline covering 400 RLBench-style Franka demonstrations across four tasks; compared adaptive and fixed-window tokenization under matched token budgets.
- Evaluated reconstruction fidelity, event-region error, and representation efficiency. Uniform chunking matched or outperformed adaptive tokenization in the evaluated setting; entropy/contact correlation was $r = 0.10$.

Robot Reliability OS | Simulation Runtime Supervision

- Built a MuJoCo pick-and-place control center with runtime risk monitoring, supervisor intervention, recovery trajectories, and episode-level telemetry.
- Implemented controlled disturbances including sensor/actuator noise, grip degradation, and obstacles; instrumented success, collision, drop, timeout, intervention, and recovery metrics.

PROFESSIONAL EXPERIENCE

Technical Lead | CitiusTech

Bengaluru, India | Oct 2024 - Present

- Lead architecture and delivery of production AI/ML systems across ingestion, feature generation, inference, evaluation, observability, and operational reliability; mentor an interdisciplinary engineering team.
- Built Python/SQL and config-driven data pipelines with canonical representations for repeatable development, dataset consistency, evaluation, and inference.
- Developed RAG-based clinical AI and claims-intelligence workflows; implemented LLM evaluation, precision/recall benchmarking, schema validation, regression testing, and stage-level health checks.

Senior Machine Learning Engineer | Microsoft

Bengaluru, India | Aug 2023 - Oct 2024

- Built production ML workflows for anomaly detection, NLP, search/relevance, and graph analytics; designed Kubernetes and Kubeflow training and deployment pipelines.
- Ran distributed training and large-scale data experiments using Python, deep-learning frameworks, Transformers, and Spark.
- Developed monitoring and calibration approaches that reduced false positives by 30%; built reliability workflows that reduced mean time to detection from 14 days to under 48 hours across four product teams.

Principal Analyst | AB InBev

Bengaluru, India | Oct 2022 - May 2023

- Developed predictive modeling, forecasting, and Isolation Forest anomaly-detection workflows; built quantitative models and executive analytics supporting \$10M+ strategic investment decisions.

Backend Developer | Panth.App

Bengaluru, India | Mar 2022 - Aug 2022

- Built computer-vision pipelines with OpenCV and CNNs for robotics-oriented workloads; optimized inference and deployment with OpenVINO, TFLite, and TVM under edge constraints.

Software Engineer and Developer | Ederra Technologies / Vikramaditya Enterprises LLP

Bengaluru, India | Jun 2017 - Nov 2021

- Developed production web applications, backend workflows, and Python/NLP data-processing pipelines using BERT/Transformers, APIs, and data-mining techniques.

ADDITIONAL AI PROJECTS

SceneSenseAI Multimodal perception and contextual narration combining YOLOv8 visual detection with language-model reasoning; added evaluation and observability to track quality and failure modes.

BreakBot Local-first security audit workbench with controlled probing, human review, evidence redaction, and reviewed report exports for authorized applications.

EDUCATION

Doctor of Business Administration in AI and ML Walsh College | 2025 - Present. Dissertation: POLARIS cross-embodiment robot policy transfer.

Postgraduate Program in AI and ML McCombs School of Business | 2021 - 2023.

Bachelor of Computer Applications Indian University for Peace and Education | 2014 - 2017.

CERTIFICATIONS AND RESEARCH INTERESTS

Microsoft Azure Solutions Architect Expert (AZ-305) | Azure Administrator (AZ-104) | Azure Fundamentals (AZ-900)

Research interests: vision-language-action and robot foundation models, imitation learning, world models, learned dynamics, and event-aware trajectory representations.